4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.1. Size, living, habitat, living and niche:

- This phylum is one of the largest phyla of the Animal Kingdom.
- The majority of nematodes are free-living forms universally present in enormous numbers in all types of environments or habitats, including the sea bottom, bottoms of all kinds of freshwater bodies, and the surface layer of all kinds of soils.
- There are also many parasitic forms, which attack all kinds of plants and animals.
- As parasites of animals, many of them are found in the alimentary canal, liver, kidneys, lungs, blood, lymph, etc.
- The numerous species, which infect crop plants, domestic animals and man, make of this phylum one of man's Worst enemies.
- Despite their great abundance and wide distribution, nematodes are amongst the least conspicuous animals because the free-living forms and plant parasites are mostly less than 1 mm in length and many are microscopic.
- Animal parasites although large, they are all endoparasites hidden of sight.

2.2. *Ascaris lumbricoides*:

2.2.1. Name, niche, habitat:

- *A. lumbricoides* is commonly named the large intestinal round worm of man.
- It is usually found in the lumen of the intestinal tract of man in all parts of the world except in the cold, dry climates.
- Although the highest frequency occurs in tropical areas, yet this infection is also common in many temperate regions of the world.
- It is one of the important intestinal parasites of man in Egypt.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.2. Morphology:

- Females (20-35 cm) are longer than the males (15-30 cm), the latter are also distinguished by a curved posterior end bearing a slit-like cloacal opening from which protrudes a pair of copulatory spicules.
- The position of the excretory canals is marked by two broad brownish lateral lines, while whitish dorsal and ventral lines similarly mark the position of the main nerve cords.
- The mouth opening lies at the anterior end of the worm, and the excretory pore is located on the ventral surface about 2 mm behind the anterior tip.
- In the female, the genital opening is situated ventrally at the end of the first third of the body and the anus is located very close to the posterior extremity of the worm.
- In the male, there is a sub terminal cloacal opening.

2.2.3. Muscular system:

- The musculature consists excusively of longitudinal fibers, circular muscles are completely lacking.
- Since the lateral cords divide the muscles into dorsal and ventral regions, Asians and most other nematodes move by dorsoventral and not by lateral bending movements.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.4. Structure of the body wall:

- *Ascaris* appears circular in cross section.
- The body wall is covered by d layer of epidermis which secretes an outer layer, the cuticle, and is followed by a muscular layer.
- Unlike that of insects the cuticle of Ascaris is not chitinous.
- The epidermis consists essentially of a thin syncytial layer which bulges inwards to form four longitudinal ridges (cords) a dorsal and ventral ridges and two lateral ridges.
- The dorsal and ventral ridges enclose the dorsal and ventral nerve cords respectively while the two lateral ridges enclose the two excretory canals.
- The muscle layer is built up exclusively of longitudinal muscle fibers.
- These are arranged into four quadrants, each quadrant including about 150 muscle cells.
- Each muscle cell is differentiated into a fibrillar or contractile outer portion and an inner cytoplasmic portion.
- The cytoplasınic portion includes the nucleus and is drawn out into cytoplasmic processes, which are directed towards either the dorsal or ventral lines.
- Between the body wall and the internal organs is a wide perivisceral cavity called the pseudocoelom or pseudocoel.
- This cavity is formed by the union of large vacuoles included in 2-5 very large parenchyma cells.
- The pseudocoel contains a fluid, which has a relatively high hydrostatic pressure

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.5. Digestive system and nutrition:

- The mouth is surrounded by three cuticular lips, one of them is dorsal while the other two are ventrolateral.
- The alimentary canal is a simple straight tube that is differentiated into the following parts:
- a) The stomodaeum, includes a short buccal cavity followed by a long muscular esophagus.
- Both structures are lined with a cuticular layer that is continuous with the cuticle on the surface of the body wall.
- b) The midgut or intestine, a straight and long tube which does not show distinct differentiation into regions. Epithelial cells, which line the intestine, have brush borders.
- c) The proctodaeum, a flattened tube that, like the stomodaeum, is lined with a cuticular layer that is continuous with the cuticle of the body wall.
- In the female worm, this tube opens to the exterior by the anal opening, while in the male it joins the genital duct into the cloaca, which opens by the cloacal opening.
- Adult *Ascaris* feeds on semi-digested food and debris, which are found in the intestine of the host.
- The following enzymnes have been demonstrated in the gut of Ascaris: amylase, maltase, protease, peptidase, lipase and esterase.
- This indicates that extra cellular digestion of the ingested food material occurs in the gut of the worm.
- The digested food is then absorbed by the intestinal cells from which it passes by diffusion into the perivisceral fluid, to the different parts of the body.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:
2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.6. Respiration:

- Since Ascaris worms live in the lumen of the intestine, they have access to little oxygen and accordingly they respire anaerobically.
- However, these worms can utilize oxygen if it is offered to them in vitro, i.e. they are facuitative rather than obligate anaerobes.

2.2.7. Excretory system:

- This is formed from a single H-shaped system of tubules, which open, by a mid-ventral excretory pore close behind the mouth.
- There are no flame cells or other current-producing mechanisms, the excretory products and excess water are forced through the excretory pore by the hydrostatic pressure exerted by the perivisceral fluid on the excretory canals of the moving worms.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:

II. THE TRIPLOBLASTICA:

II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides* (cont.):

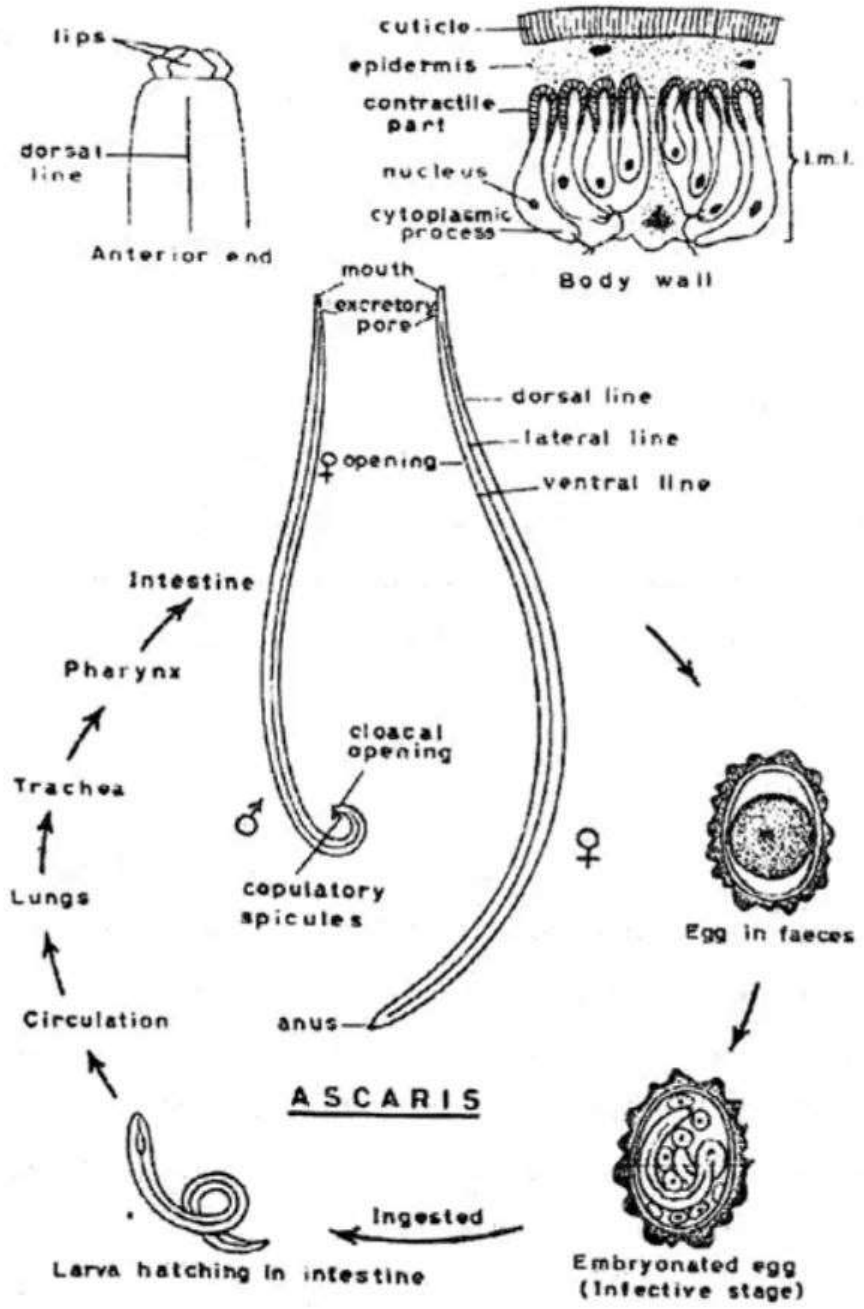

Fig. 40: Morphology, body wall, digestive system and life cycle of *Ascaris sp.*

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.8. Nervous system:

- This consists of a ganglionic nerve ring that is found around the esophagus.
- From this ring arises six nerve cords, the largest and longest of which are two, dorsal and a ventral nerve cord running along the dorsal and ventral lines respectiveiy. These two cords are interconnected at intervals by a number of transverse connectives.
- There are certain sensory papillae, which are arranged on the lips as well as at the posterior end of the worm.

2.2.9. Reproductive system and reproduetion:

- In the male, there is a single thread-like and coiled testis.
- It leads to a vas deferens that is thicker than the testis, and then to a wider tube, the vesicula semninalis.
- This ends in a short narrow muscular tube, the ejaculatory duct, which joins the proctedaeum into the cloaca.
- Dorsal to the proctodacum there is a pair of muscular sacs containing setae, which can be protruded out through the cloaca during copulation.
- The setae are known as the copulatory spicules, while the sacs are called the penial sacs.
- The anterior end of the testis is glandular and is believed to be the part that produces the spermatozoa.
- In the female, there are two ovaries, which are thread-like, convoluted tubules, one of them is situated anteriorly while the other is situated posteriorly.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:
2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.9. Reproductive system and reproduetion (cont.):

- Each ovary leads into an oviduct that passes into a wider tube, the uterus.
- The two uteri join together to form a thin tubular vagina that opens at the end of the anterior third of the body on the ventral surface.
- Each ovary is made up of a continuous sac consisting of an epihelial iayer and a germinal rachis.
- The germinal rachis contains maturing cells.
- During copulation, the male meets the female worm in such a way that the cloacal opening of the male faces the vaginal opening of the female, this being aided by the copulatory spicules of the male worm.
- The spermatozoa travel down the vagina to reach the lower parts of the uteri where fertilization of the ova occurs.
- The produced zygote is then enveloped with a vitelline membrane (sometimes called the fertilization membrane), a chitinous sheli, and an external protein coat known as the mammilated coat.
- The vitelline membrane and the chitinous shell are both secreted by the zygote, whereas the mammillated layer is secreted by certain gland cells in the wall of the uterus.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:

II. THE TRIPLOBLASTICA:

II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.10. Life cycle:

- The eggs passed in the faeces of the human host include an uncleaved zygote.
- The eggs are broadly ovoid, measuring 0.06-0.08 x 0.04-0.05 mm.
- A single female worm usually produces about 200.000 eggs / day on the average.
- Further development of the passed eggs is largely dependent on the availability of oxygen, moisture and the temperature in the outer environment.
- In a very cold and dry atmosphere, it is believed that the eggs may remain inactive, but alive, for six years.
- At a temperature of 22-35°C and in the presence of sufficient oxygen and a certain degree of humidity, the zygote develops within 9-13 days into a coiled rhabditiform larva.
- After another week, the rhabditiformn larva undergoes the first moult to form the second-stage rhabditiform larva while still within the original egg envelopes, and this is the infective stage of the parasite.
- Man acquires the infection by ingesting eggs containing the second-stage rhabditiform larva.
- In his intestine, the egg envelopes are digested and the larvae are released.
- The released larvae burrow into the intestinal mucosa, penetrate a blood vessel and are then carried to the liver via the hepatic portal vein.
- There, they may remain for a few days, but they continue traveling to the heart, and thence من هناك via the pulmonary arteries to the lungs.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:
2. PHYLUM ASCHELMINTHES (Nematodes)::
2.2. *Ascaris lumbricoides*:

2.2.10. Life cycle (cont.):

- Within the lungs the larvae moult twice, break out of th capillaries into the alveoli, and are finally driven by the cılia linin the bronchi and trachea to the pharynx, eventually reaching th small intestine.
- There, they undergo their final moult and become mature, som 8-9 weeks after the initial ingestion of the eggs.
- It is important to note that the tissue-migrating larvae of *Ascar* ingest sufficient amounts of blood to satisfy their nutrition requirements up to the fourth moult.
- It is significant that the most rapid growth phase does not tak place until the intestine, with its abundant supply of foo materials, is reached.

4. SUBKINGDOM METAZOA

4.3. Classification of Metazoa:
II. THE TRIPLOBLASTICA:
II.I. THE ACOELOMATA:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.2. *Ascaris lumbricoides*:

2.2.11. Host-parasite relationships:

- a) Effect of migrating larvae on the host:
- The migrating tissue stages of *Ascaris* produce little effects when they pass from the intestine to the liver and then to the heart.
- The presence of the larvae in the lungs, usually initiate certain tissue reactions there.
- When the larvae break out the pulmonary capillaries to reach the alveoli, they cause minute hemorrhages, and these in heavy infections may allow for secondary infections of the lungs with certain pathogenic bacteria.
- b) Effect of the adult worms on the host:
- In light infections, there may be no apparent pathological changes. However, in may lead to serious problems.
- *Ascaris*-infected school children are usually shorter and have less memory and thinking capacity than the uninfected ones.
- This is partly attributed to the fact that the worms secrete certain substances, which inactivate trypsin, thus the host's food proteins are left undigested in the lumen of the host's intestine.
- Under certain conditions, the numerous worms found in heavy infections, which frequently occur in children, nutritional demands and the space requirements of the parasite infections may be entangled together in masses, which lead to intestinal problem.

4. SUBKINGDOM METAZOA

ter:
Metazoa:
yncytial theory: 2. The colonial theory: 3. The polyphyletic theory:
n of Metazoa:
OBLASTICA
1. PHYLUM COELENTERATA
1.1. Habitata:
1.2. Classification:
A. Class Hydrozoa (Hydra-like animals): B. Class Scyphozoa (Cup-like animals).
C. Class Actinozoa (=Anthozoa).
1 - Order Alcyonaria (or Octocorallia): 2- Order Zoantheria (or Hexacorallia):
PLOBLASTICA
A. Acoelomata
1. Phylum Platyhelminthes. 2. Phylum Aschelminthes.
B. Coelomata
1. Phylum Annelida. 2. Phylum Arthropoda. 3. Phylum Mollusca. 4. Phylum Echino
5. Phylum Chordata.

A. CLASS HYDROZOA
I- HYDRA
I.I. Habitat: I.II. Morphology:
I.III. Cellular structure:
a) In the Ectoderm:
1 - The myo- or Musculo-epithelial cells 2- The interstitial cells 3- The nematoblasts (o
4- The nerve cells 5- The sensory cells, re narrow columnar cells
6- The reproductive cells 7- The mucous cells
b) In the Endoderm:
1- The nutritive or musculo-nutritive cells 2- The glandular cells are granular cells
I.IV. Locomotion:
1- Gliding. 2- Walking. 3- Somersaulting. 4- Floating.
I.V. Nutrition:
I.VI. Respiration and Excretion:
I.VII. Sensitivity:
I.VIII. Reproduction:
1- Asexual reproduction by budding: 2- Regeneration: 3- Sexual reprod
II- OBELIA:
II.I. Habitat, forms and living:
II.II. Morphology and Cellular structure
1. The colony
a) The hydranth: b) The blastostyle:
2. The medusa
II.III. Comparison between the polyp and the medusa:
SS SCYPHOZOA:
B.1. AURELIA:
B.1.1. Habitat and medusoid form: B.1.2. Morphology: B.1.3. Cellular structure:
B.1.4. Locomotion: B.1.5. Nutrition: B.1.6. Respiration and Excretion: B.1.7. Sensitivity:
B.1.8. Reproduction and Life cycle:
S ACTINOZOA
1. Order ALCYONARIA
1.1. ALCYONIUM
1.1.1. Form and habitat: 1.1.2. Morphology: 1.1.3. Cellular structure: 1.1.4. Nutrition:
1.1.5. Respiration and Excretion: 1.1.6. Reproduction and Life cycle:
2. Order ZOANTHARIA
a. SEA ANEMONES
a.1. Shape, color, habitat and skeleton: a.2. Morphology: a.3. Cellular structure:
a.4. Nutrition: a.5. Respiration and Excretion: a.6. Reproduction and Life cycle:
b. STONY CORALS:
b.1. Habitat, exoskeletons and forms: b.2. Reproduction:
rata:

4. SUBKINGDOM METAZOA (cont.)

TRIPLOBLASTICA:

THE ACOELOMATA:

PHYLUM PLATYHELMINTHES (Flat-worms):

1.1. Main character: 1.2. Living:

1.3. Classification:

A. Class Turbellaria:

B. Class Trematoda:

1. Order Monogenea (Monogentic trematodes): 2. Order Digenea (Digenetic trematodes):

C. Class Cestoda:

A. Class Turbellaria:

A.1. PLANARIA:

A.1.1. Habitat: A.1.2. Morphology: A.1.3.Histological structure: A.1.4. Locomotion:

A.1.5. Digestive system and nutrition: A.1.6. Respiration: A.1.7. Excretion and Osmore

A.1.8. Nervous system:

A.1.9. Reproductive system and Reproduction:

1- Asexual reproduction: 2- Regeneration: 3- Sexual reproduction:

B. CLASS TREMATODA

B.1. ORDER DIGENEA (Digenetice trematodes)

B.1.1. Hosts: B.1.2. Medical veterinary and attention:

B.1.3. Niches:

B.1.3.1. Liver flukes (Fasciola):

B.1.3.1.1. Hostes: B.1.3.1.2. Species, distribution and effects:

B.1.3.1.3. Fasciola gigantica:

B.1.3.1.3.1. Morphology: B.1.3.1.3.2. Histological structure:

B.1.3.1.3.3. Muscular system and locomotion: B.1.3.1.3.4. Digestive system

B.1.3.1.3.5. Respiration: B.1.3.1.3.6. Exeretory system and excretion:

B.1.3.1.3.7. Nervous system: B.1.3.1.3.8. Reproductive system and reprodu

B.1.3.1.3.9. Life cycle:

B.1.3.1.3.10. Host-parasite relationships:

a) Efect of the parasite on the intermediate host: b) Effect of the parasite on the definitive host:

B.1.3.2. Blood flukes (SCHISTOSOMA SP.)

B.1.3.2.1. Niche, hosts sexes and name: B.1.3.2.2. Disease, infected individuals and species:

B.1.3.2.3. Schistosoma haenatobium & Schistosoma mansoni:

B.1.3.2.3.1. Morphology: B.1.3.2.3.2. Digestive System and Nutrition: B.1.3.2.3.3.

B.1.3.2.3.4. Excretory, Nervous and Muscular systems:

B.1.3.2.3.5. Reproductive system and Reproduction: B.1.3.2.3.6. Life cycle:

B.1.3.2.3.7. Host-parasite relationships:

B.1.3.3. Intestinal Flukes (Heterophyes heterophyes):

B.1.3.3.1. Niche and hostes: B.1.3.3.2. Morphology: B.1.3.3.3. Life cycle:

B.1.3.3.4. Host-parasite relationships:

C. CLASS CESTODA (Cestodes or Tapeworms):

C.1. Hostes: C.2. Niche: C.3. General structure:

C.4. *Taenia saginata* and *Taenia solium*:

C.4.1. Hostes and niche: C.4.2. Morphology: C.4.3. Structure of the body wall: C.4.4. Musc

C.4.5. Nutrition: C.4.6. Respiration: C.4.7. Excretory system and excretion: C.4.8. Nerv

C.4.9. Reproductive system and reproduction: C.4.10. Life cycle: C.4.11. Host-parasite relatio

C.5. Modifications correlated with parasitic life in trematodes and cestodes:

1.4. General Characters of Platyhelminthes:

2. PHYLUM ASCHELMINTHES (Nematodes)::

2.1. Size, living, habitat, living and niche:

2.2. *Ascaris lumbricoides*:

2.2.1. Name, niche, habitat: 2.2.2. Morphology: 2.2.3. Muscular system: 2.2.4. Structure of the body wall: 2.2.5. Digestive system

Contents

ON
ition of Systematic Zoology:
ers:
sity:
ification of living organisms:
Subkingdom Protozoa: II. Subkingdom Parazoa: III. Subkingdom: Metazoa:
M PROTOZOA (PHYLUM PROTOZOA)
es numbers:
ification:
Subphylum Mastigophora
1. Movement: 2. Habitats: 3. Main character: 4. Classification:
5. EUGLENA SP.:
5.1. Habitat: 5.2. Morphology: 5.3. Locomotion: 5.4. Nutrition: 5.5. Osmoregulation
5.6. Respiration and Excretion: 5.7. Reproduction: 5.8. Sensitivity: 5.9. Is Euglena a plant or an animal?
6. TRYPANOSOMA SP.:
6.1. Habitat: 6.2. Pathogenicity: 6.3. Common species: 6.4. Morphology: 6.5. Locomotion:
6.6. Nutrition: 6.7. Osmoregulation: 6.8. Respiration and Excretion: 6.9. Reproduction:
6.10. Trypanosoma gambiense and Trypansoma rhodesiense:
6.10.1. Diseases: 6.10.2. Life cycle: 6.10.3. Host-parasite relationships:
6.11. Trypanosoma lewisi:
6.11.1. Host: 6.11.2. Life history:
6.12. LEISHMANIA SP.:
6.12.1. Host: 6.12.2. Diseases: 6.12.3. Life history: 6.12.4. Species:
6.12.5. Intermediate hosts:
. Subphylum Sarcodina
1. Movement: 2. Living:
3. Common examples:
3.1. AMOEBA SP.:
3.1.1. Habitat: 3.1.2. Morphology 3.1.3. Locomotion: 3.1.4. Nutrition: 3.1.5. Osmoregulati
3.1.6. Respiration and Excretion: 3.1.7. Reproduction: 3.1.8. Encystment: 3.1.9. Sensitivity:
3.2. ENTAMOEBA SP.:
3.2.1. Its similarity and difference to Amoeba sp.:
3.2.2. Species:
3.2.2.1. Entamoeba histolytica:
3.2.2.1.1. Niche: 3.2.2.1.2. Stages: 3.2.2.1.3. Morphology and life cycle:
3.2.2.1.4. Host-Parasite relationships:
3.2.2.2. Entamoeba coli:
3.2.2.2.1. Niche: 3.2.2.2.2. Life cycle: 3.2.2.3. Entamoeba gingivalis: 3.2.2.3.1. Niche:
3.2.2.3.2. Life cycle:
3.3. FORAMINIFERA:
3.3.1. Habitat: 3.3.2. Size: 3.3.3. Important: 3.3.4. Morphology: 3.3.5. Vegetative ac
3.3.6. Reproduction:
I. Subphylum Sporozoa
1. Movement: 2. Live cycle: 3. Habitat: 4. Characters:
5. Common examples:
5.1. MONOCYSTIS:
5.1.1. Niche: 5.1.2. Morphology: 5.1.3. Vegetative activities:
5.1.4. Respiration and excretion: 5.1.5. Reproduction and life cycle:
5.2. PLASMODIUM SP.:
5.2.1. Species: 5.2.2. Malaria:
5.2.3. Life cycle phases:
a) Fixed tissue phase b) Erythrocytic phase c) Invertebrate phase
5.2.4. Host-parasite relationships:
. Subphylum Ciliophora
1. Characters:
2. Common species:
2.1. PARAMECIUM SP.:
2.1.1. Habitat: 2.1.3. Locomotion: 2.1.4. Reproduction: 2.1.5. Sensitivity:
2.2. VORTICELLA SP.
2.2.1. Habitat: 2.2.2. Morphology: 2.2.3. Vegetative activities:
2.2.4. Osmoreguiation, respiration and excretion: 2.2.5. Reproduction:
2.3. BALANTIDIUM SP.:
2.3.1. Hosts:
2.3.2. Balantidium coli:
2.3.2.1. Host and distribution: 2.3.2.2. Niche: 2.3.2.3. Stages: 2.3.2.4. Morphology
2.3.2.5. Nutrition: 2.3.2.6. Morphology, nutrition and life cycle: 2.3.2.7. Host-parasite relationships:
ral characters of Protozoa:

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اخر محاضرة زولوجي Z102

د بهاء قناوي

دعواتكم بالتوفيق لكل الطلاب

By\Ahmed Rabie